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Amended Set of Claims (marked-up)

1. A method of predicting a likelihood of survival of a patient in need of chimeric antigen receptor (CAR) T cell therapy comprising:

measuring a level of CD27+CD28+ naïve Th cells and a level of intermediate monocytes in an apheresis product from said patient; and

determining said likelihood of survival of said patient at least in part from said level of CD27+CD28+ naïve Th cells and said level of intermediate monocytes in said apheresis product,

wherein said patient is determined to have an increased likelihood of survival if the level of CD27+CD28+ naïve Th cells is over a cut-off percentage value measured as a percentage of total leukocytes~~,~~ and the level of intermediate monocytes is below a cut-off percentage value measured as a percentage of total leukocytes, and

wherein said patient is determined to have a decreased likelihood of survival if the level of CD27+CD28+ naïve Th cells is below said cut-off percentage value and the level of intermediate monocytes is above said cut-off percentage value.

2. The method of claim 1, wherein said cut-off percentage value of CD27+CD28+ naïve Th cells measured as a percentage of total leukocytes is around 0-0.1%, 0.1%-0.5%, 0.5%-1.0%, 1.0-5%, 5-10%, 10-15%, 10-20%, 20-30%, 30-40%, 40-50%, or more preferably around 0.27%; and wherein said cut-off percentage value of intermediate monocytes measured as a percentage of total leukocytes is around 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, or 20%, preferably between 1 and 5%, and even more preferably below around 3%.

3. The method of claim 1, further comprising:

~~measuring a level of intermediate monocytes in said apheresis product from said patient; and~~

~~determining said likelihood of survival of said patient at least in part from said level of intermediate monocytes in said apheresis product,~~

~~wherein said patient is determined to have an increased likelihood of survival if the level intermediate monocytes is below a cut-off percentage value measured as a percentage of total leukocytes, and wherein said patient is determined to have a decreased likelihood of survival if the level of intermediate monocytes is above said cut-off percentage value.~~

~~4. The method of claim 3, wherein said cut-off percentage value is around 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, or 20%, preferably between 1 and 5%, and even more preferably below around 3%.~~

~~5. The method of claim 1, further comprising:~~

measuring a level of CD27-CD28+ TEMRA Treg cells in said apheresis product from said patient; and

determining said likelihood of survival of said patient also~~at least in part~~ from said level of CD27-CD28+ TEMRA Treg cells in said apheresis product,

wherein said patient is further determined to have an increased likelihood of survival if the level CD27-CD28+ TEMRA Treg cells is above a cut-off percentage value measured as a percentage of total leukocytes, and wherein said patient is further determined to have a decreased likelihood of survival if the level of CD27-CD28+ TEMRA Treg cells is below said cut-off percentage value, preferably~~.~~

~~6. The method of claim 5,~~ wherein said cut-off percentage value of CD27-CD28+ TEMRA Treg cells measured as a percentage of total leukocytes is around 0.1%, 0.2%, 0.3%, 0.4%, 0.5%, 0.6%, 0.7%, 0.8%, 0.9%, 1%, 1-5%, 5-10%, 10-20%, preferably between 0.05-0.2%, 0.2-0.25%, 0.25-0.5%, 0.5-0.6%, 0.6-0.7%, 0.7-0.8%, 0.8-0.9%, 0.9-1%, 1-5%, 5-10%, 10-15%, and more preferably around 0.1705%~~;~~

and/or

~~7. The method of claim 1, further comprising:~~

measuring a lymphocyte to leukocyte ratio in a baseline hematology count of said patient; and

determining said likelihood of survival of said patient also~~at least in part~~ from said lymphocyte to leukocyte ratio,

wherein said patient is further determined to have an increased likelihood of survival if the lymphocyte to leukocyte ratio is above a cut-off value, and wherein said patient is further

determined to have a decreased likelihood of survival if the lymphocyte to leukocyte ratio is below said cut-off value, preferably.

8. The method of claim 7, wherein said cut-off value of the lymphocyte to leukocyte ratio is 1%, 1-5%, 5-10%, 10-15%, 15-20%, 20-25%, 25-30%, 30-35%, and preferably below 5.2%; and/or

9. The method of claim 1, further comprising:

measuring a lymphocyte to monocyte ratio in a baseline hematology count of said patient; and

determining said likelihood of survival of said patient alsoat least in part from said lymphocyte to monocyte ratio,

wherein said patient is further determined to have an increased likelihood of survival if the lymphocyte to monocyte ratio is above a cut-off value, and wherein said patient is further determined to have a decreased likelihood of survival if the lymphocyte to monocyte ratio is below said cut-off value, preferably.

10. The method of claim 9, wherein said cut-off value of the lymphocyte to monocyte ratio is between 0 and 0.5, 0.5-1.0, 1.0-1.5, 1.5-2.0, 2-5, 5-10, 10-15, and preferably 0.79.

411. A method for manufacturing an immunotherapy product comprising:

(a) preparing an apheresis product from a blood sample from a subject;

(b) measuring a level of CD27+CD28+ naïve Th cells and measuring a level of intermediate monocytes in said apheresis product; and

(c) increasing an amount of CD27+CD28+ naïve Th cells collected for processing if said level of CD27+CD28+ naïve Th cells in said apheresis product is below a cut-off percentage value measured as a percentage of total leukocytes in said apheresis product and decreasing the level of intermediate monocytes in said apheresis product prior to further processing if said level of intermediate monocytes in said apheresis product is above a cut-off percentage value measured as a percentage of total leukocytes in said apheresis product.

512. The method of claim 411, wherein said cut-off percentage value of CD27+CD28+ naïve Th cells measured as a percentage of total leukocytes is around 0-0.1%, 0.1%-0.5%, 0.5%-1.0%, 1.0-5%, 5-10%, 10-15%, 10-20%, 20-30%, 30-40%, 40-50%, or more preferably around

0.27%, and wherein said cut-off percentage value of intermediate monocytes measured as a percentage of total leukocytes is around 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, or 20%, preferably between 1 and 5%, and even more preferably around 3%.

613. The method of claim 411, further comprising:

~~measuring a level of intermediate monocytes in said apheresis product; and~~

~~decreasing the level of intermediate monocytes in said apheresis product prior to further processing if said level of intermediate monocytes in said apheresis product is above a cut-off percentage value measured as a percentage of total leukocytes in said apheresis product.~~

~~14. The method of claim 13, wherein said cut-off percentage value is around 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, or 20%, preferably between 1 and 5%, and even more preferably around 3%.~~

~~15. The method of claim 11, further comprising:~~

~~–~~measuring a level of CD27-CD28+ TEMRA Treg cells in said apheresis product; and

~~–~~increasing an amount of CD27-CD28+ TEMRA Treg cells collected for processing if said level of CD27-CD28+ TEMRA Treg cells in said apheresis product is below a cut-off percentage value measured as a percentage of total leukocytes in said apheresis product,

preferably~~.~~

~~16. The method of claim 15,~~ wherein said cut-off percentage value of CD27-CD28+ TEMRA Treg cells measured as a percentage of total leukocytes is around 0.1%, 0.2%, 0.3%, 0.4%, 0.5%, 0.6%, 0.7%, 0.8%, 0.9%, 1%, 1-5%, 5-10%, 10-20%, preferably between 0.05-0.2%, 0.2-0.25%, 0.25-0.5%, 0.5-0.6%, 0.6-0.7%, 0.7-0.8%, 0.8-0.9%, 0.9-1%, 1-5%, 5-10%, 10-15%, and more preferably around 0.1705%.

717. An effective dose of T cells comprising a chimeric receptor for use in a method for treating a malignancy in a patient, said method comprising:

(a) measuring a level of CD27+CD28+ naïve Th cells and a level of intermediate monocytes in an apheresis product from said patient;

(b) determining whether said patient should be administered saidan~~~~ effective dose of T cells comprising a chimeric receptor, or saidan~~~~ effective dose of T cells comprising a

chimeric receptor and a combination therapy at least in part from said level of CD27+CD28+ naïve Th cells and intermediate monocytes in said apheresis product; and

(c) administering said effective dose of T cells comprising a chimeric receptor, or said effective dose of T cells and said combination therapy based on said determining step,

wherein said patient is administered said effective dose of T cells comprising a chimeric receptor if the level of CD27+CD28+ naïve Th cells is over a cut-off percentage value measured as a percentage of total leukocytes and the level of intermediate monocytes is below a cut-off percentage value measured as a percentage of total leukocytes, and

wherein said patient is administered said effective dose of T cells comprising a chimeric receptor and said combination therapy if the level of CD27+CD28+ naïve Th cells is below said cut-off percentage value and the level of intermediate monocytes is above said cut-off percentage value.

8~~18~~. The effective dose of T cells for use~~method~~ of claim 7~~17~~, wherein said cut-off percentage value of CD27+CD28+ naïve Th cells measured as a percentage of total leukocytes is around 0-0.1%, 0.1%-0.5%, 0.5%-1.0%, 1.0-5%, 5-10%, 10-15%, 10-20%, 20-30%, 30-40%, 40-50%, or more preferably around 0.27% and wherein said cut-off percentage value of intermediate monocytes measured as a percentage of total leukocytes is around 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, or 20%, preferably between 1 and 5%, and even more preferably around 3%.

~~19. The method of claim 17, further comprising:~~

~~measuring a level of intermediate monocytes in said apheresis product from said patient;~~

~~determining whether said patient should be administered an effective dose of T cells comprising a chimeric receptor, or an effective dose of T cells comprising a chimeric receptor and a combination therapy at least in part from said level of intermediate monocytes in said apheresis product; and~~

~~administering said effective dose of T cells comprising a chimeric receptor, or said effective dose of T cells and said combination therapy based on said determining step,~~

~~wherein said patient is administered said effective dose of T cells comprising a chimeric receptor if the level of intermediate monocytes is below a cut-off percentage value measured as a percentage of total leukocytes, and wherein said patient is administered said effective dose of T cells comprising a chimeric receptor and said combination therapy if the level of intermediate monocytes is above said cut-off percentage value.~~

~~20. The method of claim 19, wherein said cut-off percentage value is around 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, or 20%, preferably between 1 and 5%, and even more preferably around 3%.~~

921. The effective dose of T cells for use~~method~~ of claim 717, further comprising:

measuring a level of CD27-CD28+ TEMRA Treg cells in said apheresis product from said patient;

determining whether said patient should be administered saidan effective dose of T cells comprising a chimeric receptor, or saidan effective dose of T cells comprising a chimeric receptor and a combination therapy alsoat least in part from said level of CD27-CD28+ TEMRA Treg cells in said apheresis product; and

administering said effective dose of T cells comprising a chimeric receptor, or said effective dose of T cells and said combination therapy based on said determining step,

wherein said patient is administered said effective dose of T cells comprising a chimeric receptor if the level of CD27-CD28+ TEMRA Treg cells is above a cut-off percentage value measured as a percentage of total leukocytes, and wherein said patient is administered said effective dose of T cells comprising a chimeric receptor and said combination therapy if the level of CD27-CD28+ TEMRA Treg cells is below said cut-off percentage value, preferably.

~~22. The method of claim 21,~~ wherein said cut-off percentage value of CD27-CD28+ TEMRA Treg cells measured as a percentage of total leukocytes is around 0.1%, 0.2%, 0.3%, 0.4%, 0.5%, 0.6%, 0.7%, 0.8%, 0.9%, 1%, 1-5%, 5-10%, 10-20%, preferably between 0.05-0.2%, 0.2-0.25%, 0.25-0.5%, 0.5-0.6%, 0.6-0.7%, 0.7-0.8%, 0.8-0.9%, 0.9-1%, 1-5%, 5-10%, 10-15%, and more preferably around 0.1705%.

1023. The effective dose of T cells for use~~method~~ of claim 717, further comprising:

measuring a lymphocyte to leukocyte ratio in a baseline hematology count of said patient;

determining whether said patient should be administered saidan effective dose of T cells comprising a chimeric receptor, or saidan effective dose of T cells comprising a chimeric receptor and a combination therapy alsoat least in part from said lymphocyte to leukocyte ratio; and

administering said effective dose of T cells comprising a chimeric receptor, or said effective dose of T cells and said combination therapy based on said determining step,

wherein said patient is administered said effective dose of T cells comprising a chimeric receptor if the lymphocyte to leukocyte ratio is above a cut-off value, and wherein said patient is administered said effective dose of T cells comprising a chimeric receptor and said combination therapy if the lymphocyte to leukocyte ratio is below said cut-off value, preferably.

~~24. The method of claim 23,~~ wherein said cut-off value of the lymphocyte to leukocyte ratio is 1%, 1-5%, 5-10%, 10-15%, 15-20%, 20-25%, 25-30%, 30-35%, and preferably 5.2%.

11~~25~~. The effective dose of T cells for use~~method~~ of claim 7~~17~~, further comprising:

measuring a lymphocyte to monocyte ratio in a baseline hematology count of said patient;

determining whether said patient should be administered said~~an~~ effective dose of T cells comprising a chimeric receptor, or said~~an~~ effective dose of T cells comprising a chimeric receptor and a combination therapy also~~at least in part~~ from said lymphocyte to monocyte ratio; and

administering said effective dose of T cells comprising a chimeric receptor, or said effective dose of T cells and said combination therapy based on said determining step,

wherein said patient is administered said effective dose of T cells comprising a chimeric receptor if the lymphocyte to monocyte ratio is above a cut-off value, and wherein said patient is administered said effective dose of T cells comprising a chimeric receptor and said combination therapy if the lymphocyte to monocyte ratio is below said cut-off value, preferably.

~~26. The method of claim 25,~~ wherein said cut-off value of the lymphocyte to monocyte ratio is between 0 and 0.5, 0.5-1.0, 1.0-1.5, 1.5-2.0, 2-5, 5-10, 10-15, and preferably 0.79.

12~~27~~. The effective dose of T cells for use~~methods~~ of claims 7 to 11~~17-26~~, wherein said combination therapy comprises immunotherapies, SRC kinase inhibitors, T cell bi-specific antibodies, anti-CD20 monoclonal antibody, anti-4-1BB, anti-CD47, TGF-beta inhibitors or dominant negative TGF-beta, mTOR/AKT agonists, histone deacetylase inhibitors, cyclophosphamide, fluorouracil, gemcitabine, doxorubicin, taxanes, chemo- or radio- therapies,

small molecule inhibitors, antibodies targeted towards enhancing anti-tumor immunity, or anti-inflammatory medications.